SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

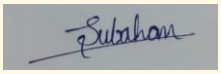
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	T. Subahan	**Reg. No.:**	192511121
Date:	12/10/26	**Duration:**	60 minutes

Code of Conduct

I T. Subahan (192511121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Vehicle Service and Maintenance Management System

1. Project Vision and Stakeholders

1.1 Project Vision

The Vehicle Service and Maintenance Management System is a software application designed to help vehicle owners book service appointments, maintain vehicle service history, receive maintenance reminders, and track the status of their vehicles during servicing. The system aims to reduce manual work, save time, improve service management, and provide a convenient platform for communication between customers and service centers.

1.2 Project Objectives

- To allow vehicle owners to book service appointments online.
- To maintain a complete history of vehicle servicing and repairs.
- To send reminders for upcoming maintenance and scheduled services.
- To enable customers to track the real-time service status of their vehicles.
- To help service center staff manage appointments and service requests efficiently.
- To improve customer satisfaction through timely updates and organized records.

1.3 Key Stakeholders

Stakeholder	Role and Responsibilities
Vehicle Owners / Customers	Book appointments, view service history, receive reminders, and track service status.
Service Center Manager	Manage service operations, staff, and appointment schedules.
Service Technicians	Perform maintenance and update service progress.
Receptionist / Service Advisor	Confirm bookings and communicate with customers.
System Administrator	Manage user accounts, permissions, and system settings.
Development Team	Design, develop, test, and maintain the application.
Product Owner	Define requirements and prioritize the product backlog.
Scrum Master	Facilitate Scrum events and remove development obstacles.

2. Create Epics and User Stories

An Epic is a large functionality that can be divided into smaller user stories. A user story describes a feature from the perspective of the person who will use it.

2.1 Identified Epics

- Epic 1: User and Vehicle Management — Registration, login, and vehicle details.
- Epic 2: Service Appointment Booking — Booking, viewing, and cancelling service appointments.
- Epic 3: Service History Management — Maintaining and viewing previous service records.
- Epic 4: Maintenance Reminders — Notifying users about upcoming services.
- Epic 5: Service Status Tracking — Tracking service progress and completion.
- Epic 6: Service Center Management — Managing appointments and updating servicerecords.

2.2 User Stories

Story ID	Epic	User Story
US01	User and Vehicle Management	As a vehicle owner, I want to register and log in, so that I can securely access my account.
US02	User and Vehicle Management	As a vehicle owner, I want to add my vehicle details, so that I can manage its service requirements.
US03	Appointment Booking	As a vehicle owner, I want to book a service appointment, so that I can get my vehicle serviced at a convenient time.
US04	Appointment Booking	As a vehicle owner, I want to view or cancel my booking, so that I can manage my service schedule.
US05	Service History	As a vehicle owner, I want to view previous service records, so that I can understand my vehicle's maintenance history.
US06	Maintenance Reminders	As a vehicle owner, I want to receive maintenance reminders, so that I can service my vehicle on time.
US07	Service Status Tracking	As a vehicle owner, I want to track the service status, so that I know whether my vehicle is being inspected, repaired, or is ready for collection.
US08	Service Center Management	As a service advisor, I want to view and manage appointments, so that I can organize the service center's daily work.
US09	Service Center Management	As a technician, I want to update the service progress, so that customers receive accurate status information.

3. Define Acceptance Criteria

Acceptance criteria specify the conditions that must be satisfied for a user story to be considered complete.

US01: User Registration and Login

- Given a new user with valid details, when they submit the registration form, then the system creates an account.
- Given a registered user enters correct login credentials, when they select Login, then the system opens their account.
- Given a user enters incorrect credentials, when they attempt to log in, then the system displays an error message.

US02: Add Vehicle Details

- Given a logged-in user, when they enter the vehicle registration number, make, model, and year, then the system saves the vehicle details.
- Given required fields are missing, when the user submits the form, then the system displays validation messages.

US03: Book a Service Appointment

- Given a registered user with a vehicle, when they select a service type, available date, and time, then the system creates a booking.
- Given a selected time slot is unavailable, when the user attempts to book it, then the system asks them to select another available slot.

- Given a booking is successful, when the system saves the appointment, then it displays a booking reference and confirmation.

US04: View or Cancel a Booking

- Given a customer has an existing appointment, when they open My Bookings, then the system displays the appointment details.
- Given a booking is eligible for cancellation, when the customer confirms cancellation, then the booking status changes to Cancelled.

US05: View Service History

- Given a vehicle has previous service records, when the owner opens Service History, then the system displays the service date, work performed, and cost when available.
- Given a vehicle has no service records, when the owner opens Service History, then the system displays a suitable message.

US06: Maintenance Reminders

- Given a vehicle has a recorded upcoming maintenance date, when the reminder date is reached, then the system generates a maintenance reminder.
- Given a reminder is generated, when the customer views notifications, then the reminder displays the vehicle and due-service information.

US07: Track Service Status

- Given a vehicle has an active service booking, when the technician updates its status, then the system displays the latest status to the customer.
- Given the service is completed, when the technician marks it as completed, then the system displays the vehicle as Ready for Collection or Completed.

US08: Manage Appointments

- Given an authorized service advisor, when they open the appointment dashboard, then the system displays scheduled service appointments.
- Given an appointment requires an update, when the advisor changes its status, then the system saves the updated information.

US09: Update Service Progress

- Given a technician is assigned to a service request, when they select a valid progress status, then the system records the update.
- Given the status is updated successfully, when the customer views the tracking page, then the latest status is displayed.

4. Prioritize the Product Backlog

The Product Backlog is an ordered list of features and requirements. The following backlog uses MoSCoW prioritization: Must Have, Should Have, and Could Have.

Priority	Story ID	User Story / Feature	Category
1	US01	User registration and login	Must Have
2	US02	Add vehicle details	Must Have
3	US03	Book service appointment	Must Have
4	US08	View and manage appointments	Must Have
5	US09	Update service progress	Must Have
6	US07	Track service status	Must Have
7	US05	View service history	Should Have

8	US04	View or cancel bookings	Should Have
9	US06	Maintenance reminders	Should Have

Reason for prioritization: User authentication and vehicle registration are foundational requirements. Appointment booking depends on the user's account and vehicle details. Service management and status tracking depend on appointments being created. Service history and reminders can be implemented after the basic service workflow is available.

5. Plan the Sprint Backlog

A Sprint Backlog contains the user stories selected for a sprint and the tasks required to complete them.

Sprint duration: 2 weeks

Sprint number: Sprint 1

Team assumption: One small Scrum team consisting of developers, a tester, a Product Owner, and a Scrum Master.

5.1 Selected User Stories for Sprint 1

Story ID	Selected User Story	Priority	Story Points
US01	User registration and login	Must Have	5
US02	Add vehicle details	Must Have	3
US03	Book service appointment	Must Have	8
	Total		16

These stories establish the minimum working flow: users can access the system, register their vehicles, and book service appointments.

5.2 Development Tasks

Story ID	Development Tasks
US01	Design registration and login screens; create the user database; implement registration and authentication; validate inputs; test valid and invalid login attempts.
US02	Design the vehicle details form; define vehicle database fields; implement add and view vehicle functions; validate required fields; test vehicle record storage.
US03	Design the appointment booking screen; create the appointment database; implement service type and date/time selection; validate availability; save booking details; display confirmation; test booking scenarios.

5.3 Sprint Activities

Day / Period	Activity
Day 1	Sprint Planning, requirement clarification, and task assignment
Days 2–4	Develop user registration and login
Days 5–6	Develop vehicle registration
Days 7–9	Develop appointment booking
Day 10	Integration testing, Sprint Review, and Sprint Retrospective

The schedule is indicative and can be adjusted according to team capacity and technical dependencies.

6. Estimate Effort Using Story Points

6.1 Estimation Technique: Planning Poker

Planning Poker is an Agile estimation technique in which team members independently estimate the relative effort of user stories. The team discusses differences and agrees on a final estimate.

The Fibonacci sequence is commonly used for story points:

1, 2, 3, 5, 8, 13, 21

Higher values represent greater effort, complexity, uncertainty, or risk.

6.2 Story Point Estimation

Story ID	Story	Story Points	Justification
US01	User registration and login	5	Requires form design, authentication logic, input validation, and security testing.
US02	Add vehicle details	3	Involves a relatively simple form, database storage, and field validation.
US03	Book service appointment	8	Requires date/time selection, availability validation, database integration, and booking confirmation.
	Total	16	Moderate workload for the planned sprint, subject to team capacity.

Justification: Vehicle registration is relatively straightforward, so it receives 3 points. Authentication requires additional validation and security considerations, so it receives 5 points. Appointment booking involves more complex scheduling and availability checks, so it receives 8 points.

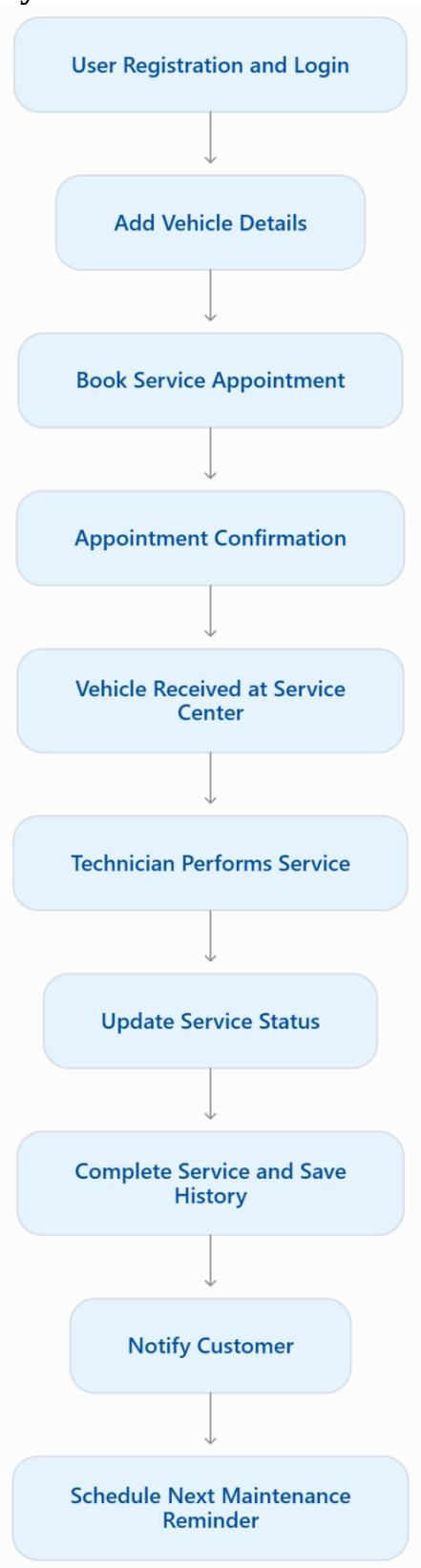
Story points represent relative effort rather than exact hours. The team should confirm these estimates through Planning Poker.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

To develop the basic functionality of the Vehicle Service and Maintenance Management System by enabling users to register and log in, add their vehicle details, and book service appointments through a simple and user-friendly interface.

System Workflow



7.2 Expected Sprint Deliverables

At the end of Sprint 1, the team expects to deliver:

1. A working user registration and login module.
2. A vehicle registration form that saves vehicle details.

3. A service appointment booking interface.
4. Appointment data stored in a database.
5. Basic validation for missing or invalid information.
6. Booking confirmation with a reference number.
7. Test cases and test results for the implemented features.
8. A demonstrable working prototype of the application's initial workflow.

7.3 Definition of Done

A user story is considered complete when:

- All acceptance criteria have been satisfied.
- The feature has been implemented and integrated.
- Required input validation is working.
- Unit and relevant integration tests have passed.
- No known critical defects remain in the feature.
- The code and functionality have been reviewed by the team.
- The feature is demonstrated during the Sprint Review.

7.4 Expected Outcome

The first sprint should produce a basic working system in which a customer can register, log in, add vehicle information, and book a service appointment. In subsequent sprints, the team can implement appointment management, service history, maintenance reminders, and service status tracking.

Conclusion

Agile Sprint Planning helps the development team divide the Vehicle Service and Maintenance Management System into manageable user stories and deliver useful features incrementally. By identifying stakeholders, creating epics, defining acceptance criteria, prioritizing the backlog, estimating story points, and planning a two-week sprint, the team can develop the system in a structured and flexible manner. Regular Sprint Reviews and Retrospectives will help improve product quality, teamwork, and customer satisfaction.